

PREVIOUS HSE QUESTIONS FROM THE CHAPTER "CHEMICAL BONDING AND MOLECULAR STRUCTURE"

1. Water is a liquid while H_2S is a gas.
 - a) Suggest the reason for the above fact. (1)
 - b) Explain the phenomenon. (2) [February 2008]
2. a) What do you understand by bond pair electrons and lone pair electrons? (2)
b) Explain the bond pair electrons and lone pair electrons in H_2O and NH_3 molecules with suitable drawings. (3) [March 2009]
3. The stability and magnetic properties of a molecule can be explained using the molecular orbital theory proposed by F. Hund and R.S. Mulliken.
 - a) Define bond order according to the M.O theory. (1)
 - b) Draw the energy level diagram for the formation of O_2 molecule. (2)
 - c) Calculate the bond order and predict the magnetic character of O_2 molecule. (2) [March 2010]
4. VSEPR theory is used to predict the shape of covalent molecules.
 - a) State the main postulates of VSEPR theory. (3)
 - b) Based on VSEPR theory predicts the shape of H_2O and NH_3 . (2) [October 2010]
5. The attractive force which holds atoms together in a molecule is called a chemical bond.
 - a) Explain the formation of a H_2 molecule on the basis of the valence bond theory (VBT). (2½)
 - b) Using the molecular orbital theory (MOT), explain why Ne_2 molecule does not exist? (1½)
 - c) Calculate the bond order of dinitrogen (N_2). (1) [March 2011]
6. a) Hydrogen bonding plays an important role in determining the physical properties of substances.
 - i) Illustrate hydrogen bonding using an example. (1½)
 - ii) Compare the boiling points of o-nitro phenol and p-nitro phenol based on hydrogen bonding. (1½)b) Describe the hybridisation and structure of PCl_5 molecule. (2) [September 2011]
7. Valence Bond Theory (VBT) and Molecular Orbital Theory (MOT) are the two important theories of chemical bonding.
 - a) Out of the following which is the hybridisation of phosphorus in PCl_5 ? (sp^3 , sp^2 , dsp^2 , sp^3d) (1)
 - b) Explain the geometry of PCl_5 molecule and account for its high reactivity. (2)
 - c) Write the molecular orbital configuration of the C_2 molecule and calculate its bond order. (2) [March 2012]
8. a) The ionic bonds have partial covalent character and the covalent bonds also show some ionic character.
 - i) Explain the covalent character of Lithium chloride using Fajan's rule (1)
 - ii) NF_3 and NH_3 show dipole moment. But the dipole moment of NF_3 is less than that of NH_3 . Why? (1)
 - iii) The covalent bond can be explained by Molecular Orbital Theory (MOT). Using MO diagram explain the paramagnetic nature of oxygen molecule. (3) [September 2012]
9. The Valence Shell Electron Pair Repulsion (VSEPR) theory helps in predicting the shapes of covalent molecules.
 - a) Arrange the bond pair electron and lone pair electron in the decreasing order of the repulsive interactions among them. (1)
 - b) A molecule of the type AB_3E_2 has three bond pairs and two lone pairs of electrons. Predict the most stable arrangement of electron pairs in this molecule. (1)